

# Interval Uploads Tab Guide

Modified on: Thu, 20 Nov, 2025 at 6:21 PM

The Intervals Uploads tab, found under the Portfolio menu, allows partners to add interval data to the Leap platform. Select grid services programs are able to utilize interval data obtained directly from approved devices or meters to assess performance and to complete partner payments. For these programs, partners will need to provide interval data through either the portal or via API. The data provided to Leap must be in the format defined below and uploaded as CSVs to the Partner Portal or SFTP server.

Interval data uploads are currently utilized in the following programs:

- **California:** Demand Side Grid Support (DSGS) program, both Option 3 and Option 4
- **Massachusetts:** Clean Peak Standard (CPS) program
- **New England:** ConnectedSolutions (CS) program
- **New York:** Special Case Resources (SCR) program, [only for SCR meters that are enrolled through Leap's meter upload process](https://support.leap.energy/support/solutions/articles/73) (<https://support.leap.energy/support/solutions/articles/73>)

(<https://developer.leap.energy/docs/upload-interval-data-sftp>), (<https://developer.leap.energy/reference/listmeterlevelintervals>).

**Automating Interval Data Uploads:** Leap provides an SFTP option for partners to automate and scale the process of sharing interval data. Learn more in our [Interval Data Upload \(SFTP\)](#)

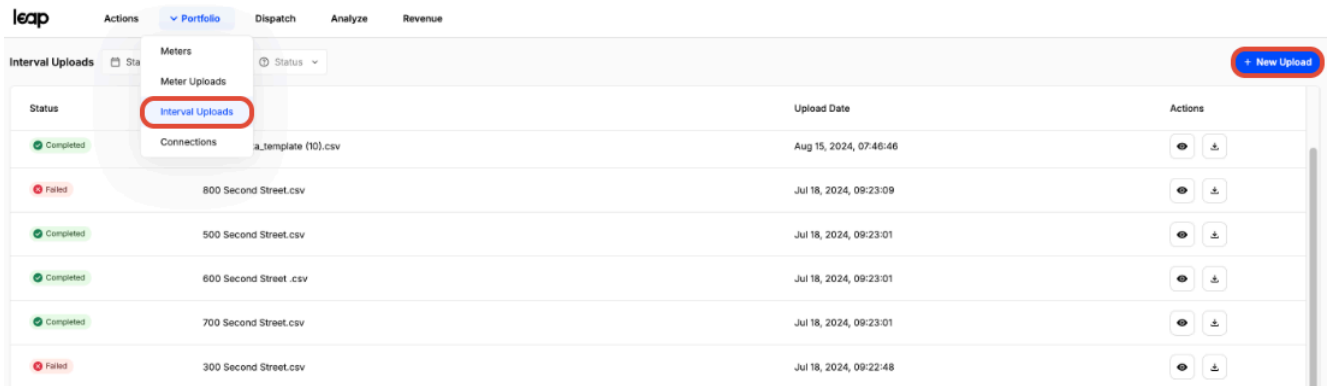
Leap strongly recommends that partners automate interval data uploads for the California DSGS and Massachusetts CPS programs. These programs require interval data submissions on

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### Uploading Interval Data to the Portal

1. From the Portfolio tab in the partner portal, expand the menu and select Interval Uploads. Select the “+ New Upload” button in the top right corner.



2. Download the template CSV file.
3. Once the CSV template has been filled out according to the specifications described below, either “drop” the completed CSV file(s) into the dialog box or select “Choose file” and then “Upload”. Multiple files may be uploaded at once.

Interval Data CSV Upload ⓘ  
Upload interval consumption data via CSV. File must be formatted according to template.  

Download template

Drag CSV files here or Choose files (max size per file 500 MB)

CancelUpload

Please note:

- It is important to carefully review all data prior to upload. All fields marked required must be included in every file.
- File specification tips:
  - File names should be unique. Leap recommends using a file naming convention such as {company\_name}\_{timestamp}.csv
  - There are no requirements for which intervals or which meters a file includes. Partners are free to structure individual files as they choose. Do note that there is a maximum file size of 500MB.

- Interval start and end time values must be converted to UTC time zone and in the ISO-8601 format. Valid formatting looks like YYYY-MM-DDTHH:MM:SSZ. Be sure to include all hours in a day in the meter's local time zone (converted to UTC time zone) versus ending the dataset in the final hours of a day in UTC time. This is particularly important when uploading data for the final day of the month or delivery period.
- All data must be provided in 15 minute intervals, and all interval start and end times must be exact to the 0:00, 0:15, 0:30, and 0:45 minute. In the ISO-8601 format, this will look like MM:SSZ as 00:00Z-15:00Z, 15:00Z-30:00Z, 30:00Z-45:00Z, 45:00Z-00:00Z. As shown, the seconds value in this ISO-8601 format must always be :00. Other interval data granularity is not supported at this time (eg. 1 minute, 5 minute, 30 minute, 1 hour).
- If there are known intervals with missing data, 'null' may be used in the energy usage fields.
  - 'Null' can be used in the 'energy\_net\_kwh' field
  - 'Null' can be used in both the 'energy\_consumed\_kwh' field and 'energy\_generated\_kwh' field in the same interval.
  - 'Null' cannot be used for only one of the two 'energy\_consumed\_kwh' or 'energy\_generated\_kwh' fields in the same interval. This will return an error.
- If duplicate data for the same interval and Meter ID is found, Leap will use data that was marked with the data field 'final' = 'true'. Otherwise the most recently uploaded data will be used and will overwrite the older, duplicate intervals.

Interval Uploads Table

All uploaded files (uploaded through both the Portal or provided by SFTP) will appear in the table on this tab. Leap processes submitted files to ensure they meet data specification and formatting requirements as described above and outlined in the data schema table below. Interval data files provided to Leap prior to April 2024 will not be displayed in the table or in API results.

A unique Job ID is generated for each submitted file and is displayed in the table.

Rows in this table will display one of the following **status options** for each Job ID:

- Processing:** Leap is still processing the data in this file; the status will update once processing is complete. Most files will process very quickly, so partners may not see the Processing status at all. Try refreshing the browser page if uploaded file(s) are not yet appearing in the table.
- Completed:** All data in the file was successfully added to the Leap platform. Clicking into the row entry in the table will display a preview of the file's contents. Interval data for associated meters is viewable in the Analyze tab as Total Load.
- Failed:** Leap rejected the file due to errors. Click into a failed row entry in the table to view detailed information about the errors found within the file. No intervals from this file were uploaded. Partners will need to correct the errors and reupload the entire file.

For each uploaded file, partners will be able to

- Download the original file:** Use the download icon under the Actions column to download the originally uploaded file.
- View further details:** Use the inspect icon under the Actions column or click on the row to view more details about the file. For successfully uploaded files that have a 'Completed' status, a preview of up to the first 100 rows of the file's contents will be displayed. For rejected files that have a 'Failed' status, detailed error information is provided.

← BackInterval Upload ErrorsElectric\_SpaceShip\_2024\_08.csvJob ID: 463a3358-6555-4747-b079-9a718460725cTotal Errors: 6Download Original

| Row Lines | Column                  | Errors                                                                                   |
|-----------|-------------------------|------------------------------------------------------------------------------------------|
| 2 - 5     | meter_id                | Meter with this meter ID was not found.                                                  |
| 2         | interval_end_time_utc   | 2024-09-01T11:05:00Z is not a valid time. The interval should align to while 15 minutes. |
| 2         | energy_generated_kwh    | -1 should not be negative.                                                               |
| 2         | energy_consumed_kwh     | -5 should not be negative.                                                               |
| 3         | interval_start_time_utc | 2024-09-01T11:05:00Z is not a valid time. The interval should align to while 15 minutes. |
| 3         | interval_end_time_utc   | 2024-09-01T11:10:00Z is not a valid time. The interval should align to while 15 minutes. |
| 4         | interval_start_time_utc | 2024-09-01T11:10:00Z is not a valid time. The interval should align to while 15 minutes. |
| 5         | interval_end_time_utc   | 2024-09-01T11:20:00Z is not a valid time. The interval should align to while 15 minutes. |

← BackInterval Upload Preview500 Second Street.csvJob ID: 39d9857-5842-42a3-a666-3c7d31a55545Total Intervals: 6Download

| meter_uuid                           | interval_start_time_utc | interval_end_time_utc | energy_net_kwh |
|--------------------------------------|-------------------------|-----------------------|----------------|
| 2fa0b3f5-050f-4590-9630-b55d4492cda3 | 2023-12-30T14:00:00Z    | 2023-12-30T14:15:00Z  | 0              |
| 2fa0b3f5-050f-4590-9630-b55d4492cda3 | 2023-12-30T14:15:00Z    | 2023-12-30T14:30:00Z  | 15             |
| 2fa0b3f5-050f-4590-9630-b55d4492cda3 | 2023-12-30T14:30:00Z    | 2023-12-30T14:45:00Z  | 347            |
| 2fa0b3f5-050f-4590-9630-b55d4492cda3 | 2023-12-30T14:45:00Z    | 2023-12-30T15:00:00Z  | 93             |
| 2fa0b3f5-050f-4590-9630-b55d4492cda3 | 2023-12-30T14:00:00Z    | 2023-12-30T14:15:00Z  | NULL           |

Interval Data Format

| Field Name              | Description                                                                                                                                         | Data Type | Required Field |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| meter_id                | Meter ID assigned by Leap. meter_id in the API                                                                                                      | string    | Yes            |
| interval_start_time_utc | The start time of the energy usage interval in UTC time zone and ISO-8601 format. Must be in 15-minute increments (00:00Z, 15:00Z, 30:00Z, 45:00Z). | datetime  | Yes            |
| interval_end_time_utc   | The end time of the energy usage interval in UTC time zone and ISO-8601 format. Must be in 15-minute increments (00:00Z, 15:00Z, 30:00Z, 45:00Z).   | datetime  | Yes            |

|                      |                                                                                                                                                                                                                                         |         |                                                                                                                                                                                |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| energy_net_kwh       | Net energy consumption amount in kWh during this interval. Calculated as net energy = energy consumed - energy generated<br><br>'null' value may be used if usage data is not available                                                 | number  | Yes, must provide one out of the three options for each interval:<br><br>1) only energy_net_kwh<br>2) both energy_consumed_kwh and energy_generated_kwh<br>3) all three fields |
| energy_consumed_kwh  | Energy consumed in kWh during this interval. No negative values.<br><br>'null' value may be used if usage data is not available                                                                                                         | number  |                                                                                                                                                                                |
| energy_generated_kwh | Energy generated in kWh during this interval. No negative values.<br><br>'null' value may be used if usage data is not available                                                                                                        | number  | For fields not provided, please leave them blank (" ") or omit the column altogether<br><br>Max accepted value of 3,000 kWh per interval                                       |
| final                | Values must be 'true' or 'false'<br><br>Indicates whether partners believe the interval data is preliminary and subject to be updated later ('false') or a final value to provide confidence in reporting and payment metrics ('true'). | boolean | No, optional                                                                                                                                                                   |
| region               | Required for electric vehicles in New England programs. Value must be an all-caps, two letter state code, eg. MA, CT<br><br>Specifies the state where the energy consumption or generation occurred during this interval.               | string  | Yes, conditional                                                                                                                                                               |

#### Sample Interval Data Entries

| meter_id                             | interval_start_time_utc | interval_end_time_utc | energy_net_kwh | energy_consumed_kwh | energy_generated_kwh | final | region |
|--------------------------------------|-------------------------|-----------------------|----------------|---------------------|----------------------|-------|--------|
| 2bcbe422-e3dc-49f3-9591-eda5f1729b86 | 2023-12-30T14:00:00Z    | 2023-12-30T14:15:00Z  | 50             | 150                 | 100                  | true  | CA     |
| 2bcbe422-e3dc-49f3-9591-eda5f1729b86 | 2023-12-30T14:15:00Z    | 2023-12-30T14:30:00Z  |                | 150                 | 100                  | true  | CA     |
| 3eetr360-t4lc-6811-2368-coe6m3429w90 | 2023-12-30T14:00:00Z    | 2023-12-30T14:15:00Z  | 50             |                     |                      | true  | CA     |

#### Interval Data Requirements and Deadlines by Program

Programs that utilize interval data directly from Partners vary in data upload frequency requirements.

- California: Demand Side Grid Support (DSGS) Option 3 program**
  - Interval data submission is required **monthly, within five business days** following the end of **each month**. Beginning in the 2025 season, if data is not provided within five business days of each month end, partners risk not meeting program requirements and Leap cannot guarantee payment for that month.
  - Interval data is required for every day (24 hours) a meter is actively enrolled in the program, not just dispatch event days or dispatch hours.
  - See the [DSGS Interval Data Deadlines](https://support.leap.energy/support/solutions/articles/73000655366-california-dsgs-interval-data-deadlines) (https://support.leap.energy/support/solutions/articles/73000655366-california-dsgs-interval-data-deadlines) article for exact monthly deadline dates.
- California: Demand Side Grid Support (DSGS) Option 4 program**
  - Interval data submission is required **within five business days of each dispatch event**. If data is not provided within five business days of each dispatch event, partners risk not meeting program requirements and receiving payment.
  - Interval data is required for every day (24 hours) a meter is actively enrolled in the program, not just dispatch event days or dispatch hours.
  - See the [DSGS Interval Data Deadlines](https://support.leap.energy/support/solutions/articles/73000655366-california-dsgs-interval-data-deadlines) (https://support.leap.energy/support/solutions/articles/73000655366-california-dsgs-interval-data-deadlines) article for exact monthly deadline dates.
- Massachusetts: Clean Peak Standard (CPS) program**
  - Interval data submission is required **monthly, within five business days** following the end of **each month**.
  - Interval data is required for every day (24 hours) a meter is actively enrolled in the program, not just dispatch event days or dispatch hours.
  - See the [CPS Interval Data Deadlines](https://support.leap.energy/support/solutions/articles/73000655802-massachusetts-clean-peak-standard-interval-data-deadlines) (https://support.leap.energy/support/solutions/articles/73000655802-massachusetts-clean-peak-standard-interval-data-deadlines) article for exact monthly deadline dates.
- New England: ConnectedSolutions (CS) program**
  - Interval data submission deadlines vary by utility.
    - For Eversource** meters: Interval data submission is required **monthly, within five business days** following the end of each month.
    - For National Grid and Rhode Island Energy** meters: Interval data submission is required **seasonally, within five business days** following the end of each annual season.
  - Interval data is required for every day (24 hours) a meter is actively enrolled in the program, not just dispatch event days or dispatch hours.
  - See the [ConnectedSolutions Interval Data Deadlines](https://support.leap.energy/support/solutions/articles/73000655801-new-england-connectedsolutions-interval-data-deadlines) (https://support.leap.energy/support/solutions/articles/73000655801-new-england-connectedsolutions-interval-data-deadlines) article for exact deadline dates.
- New York: Special Case Resources (SCR) program**
  - For only select partners that [enroll in SCR through uploading meters to Leap](https://support.leap.energy/support/solutions/articles/73000631559-adding-meters-in-new-york) (https://support.leap.energy/support/solutions/articles/73000631559-adding-meters-in-new-york) (instead of through direct customer authorization using Leap Connect), interval data at the utility meter level is required prior to enrollment AND ongoing throughout participation.

- **Historical interval data:** The last 12 months of historical interval data can be uploaded after partners have successfully created meters on the Leap platform. Complete historical interval data is a prerequisite to enroll in the program.
- **Ongoing interval data:** Submission is required **monthly, within five business days** following the end of **each month**. Interval data is required for every day (24 hours) a meter is actively enrolled in the program, not just dispatch event days or dispatch hours. See the [SCR Interval Data Deadlines](https://support.leap.energy/support/solutions/articles/73000663025-new-york-scr-interval-data-deadlines/) (<https://support.leap.energy/support/solutions/articles/73000663025-new-york-scr-interval-data-deadlines/>) article for exact monthly deadline dates.

#### **Energy Usage Data Formatting Recommendations**

To provide flexibility to partners, the interval data schema provides three options listed in the table above for how to format energy usage data. Partners may use whichever format is the best fit for their data structure and systems. Leap recommends the following depending on meter type or load type:

- **Utility or Site Meter:** Provide the 'energy\_net\_kwh' value. The single data point available should be provided in the net value field.
    - The best fit for HVAC loads is to provide the single consumption data point as the 'energy\_net\_kwh' value. There is no calculation needed to derive the net value as generation is always zero.
  - **Device:** Partners enrolled with assets at the device level likely have interval data available that is suitable for any of the formatting options.
    - For EVs, device charging is the 'energy\_consumed\_kwh' value. If the device is bidirectional, then any discharging would be the 'energy\_generated\_kwh' value. Alternatively, the single 'energy\_net\_kwh' value can be provided by calculating charging - discharging values in each interval.
    - For batteries, battery charging is the 'energy\_consumed\_kwh' value. Battery discharging or exporting is the 'energy\_generated\_kwh' value. If the battery is idle, then '0' can be used to indicate as such. If the battery is in an error state and interval data is lost, then partners should use 'null'. Alternatively, the single 'energy\_net\_kwh' value can be provided by calculating charging - discharging values in each interval.
-